

AI in Fashion Design Practice

A Review of Designer Adaptation to Artificial Intelligence

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Introduction

AI is increasingly integrated into fashion design.

Research focuses mainly on AI applications.

Limited attention to designer adaptation.

Aim: Review how designers adapt to AI and emerging Human–AI co-creativity.



Literature Review

A structured narrative review to understand designer adaptation to AI in fashion design.

- **Narrative literature review**
  Broad exploration of existing scholarship
- **~50 papers initially reviewed**
  Across databases and academic sources
- **16 key studies selected**
  Based on relevance and contribution
- **Inclusion criteria**
  Designer adaptation, AI adoption, fashion design practice
- **Thematic synthesis**
  Identification of recurring patterns and key insights

Review Process

~50
papers initially reviewed

16
key studies selected

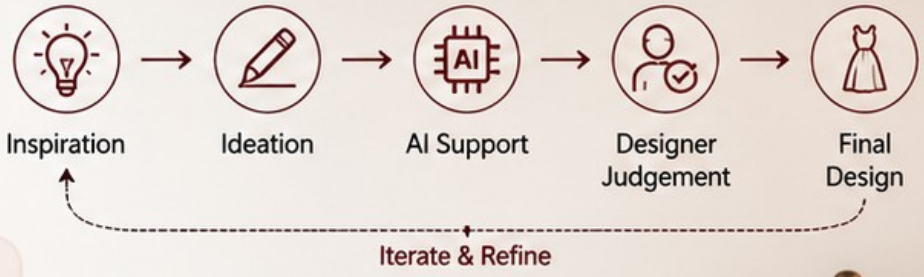
In-depth analysis

Focus Areas of Selected Studies

- Designer Adaptation
- AI Adoption
- Fashion Design Practice
- Human–AI Co-creativity

Key Findings

AI is transforming fashion design—and designers are evolving with it.



01



Transition from CAD to AI-supported design

Fashion design has evolved from CAD-based tools to intelligent AI systems that support ideation, visualisation, and decision-making.

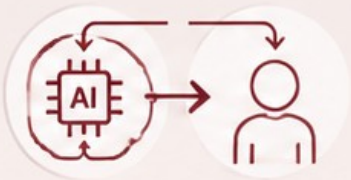
02



Designers require AI literacy and prompt engineering

New competencies are essential—understanding AI tools, crafting effective prompts, and critically evaluating AI-generated outputs.

03



AI changes workflows but designers retain creative judgement

AI enhances speed and iteration, but human creativity, aesthetic sense, and final decision-making remain irreplaceable.

04



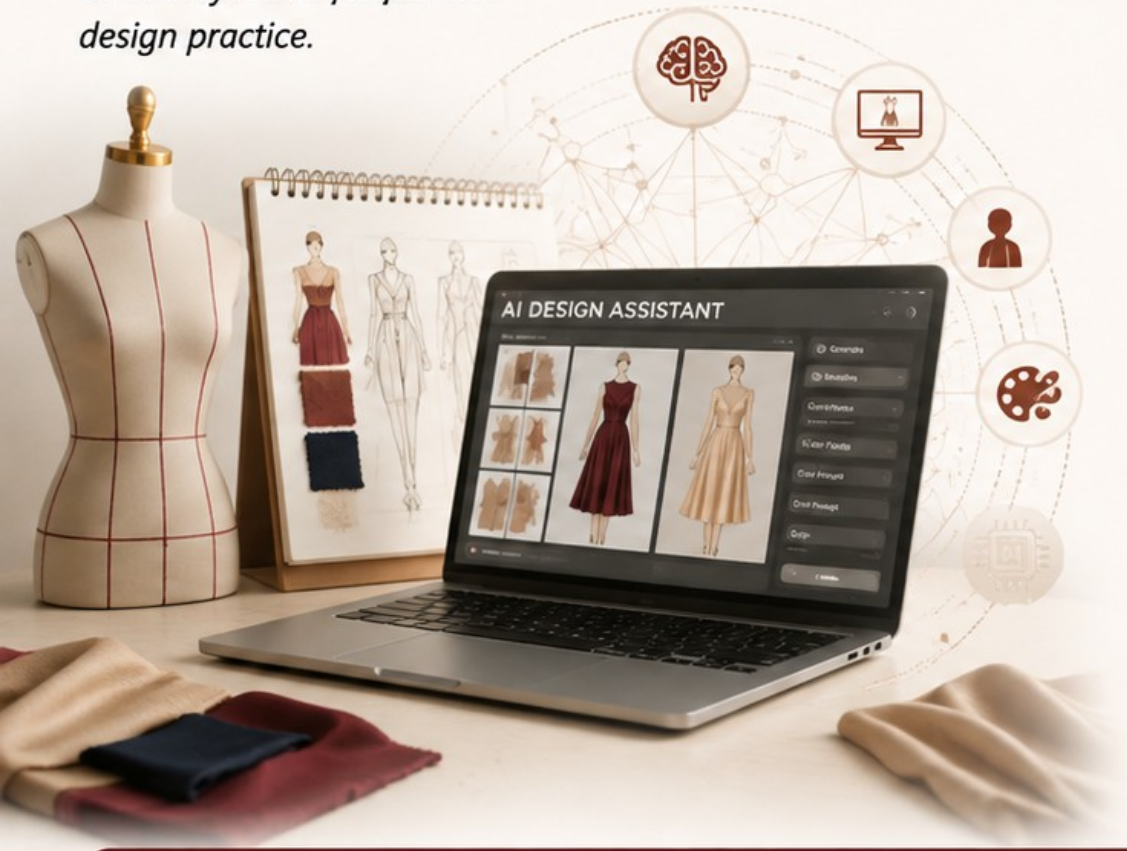
Human–AI co-creativity is emerging

A collaborative partnership is forming—where human intuition and AI capability come together to create novel and meaningful design outcomes.



Discussion

Interpreting the findings and what they mean for fashion design practice.



Successful AI adoption depends on learning and adaptation.

Designers who invest time in learning AI tools, prompt engineering, and evaluation develop confidence and integrate AI more effectively into their creative workflows.

AI supports rather than replaces designers.

AI enhances speed, exploration, and variation, but human judgement, contextual understanding, and aesthetic sensibility remain irreplaceable. AI acts as a creative partner, not a substitute.

Creativity becomes collaborative.

The most valuable outcomes emerge when human intuition, experience, and cultural insight are combined with AI's computational capabilities. Co-creation leads to richer, more innovative design solutions.

Human judgement ensures originality and authorship.

Originality is achieved through interpretation, modification, and intentional decision-making. Designers retain authorship by guiding, selecting, and refining AI-generated outputs.

Education must include AI literacy, ethics, and critical evaluation.

Fashion design education should equip designers with technical skills, ethical awareness, and critical thinking to use AI responsibly and transparently in creative practice.

Key Takeaway

AI integration in fashion design is both technological and human.
The future lies in responsible Human–AI collaboration, where technology empowers designers and human creativity leads to meaningful, sustainable outcomes.

Discussion

How the literature explains the evolution toward AI-supported fashion design.



The literature shows a clear progression that leads to **Human–AI co-creativity** in fashion design.

1 Evolution of Design Workflow



From execution → collaboration

- Manual design evolved into CAD-supported workflows.
- AI extends this further by supporting ideation, exploration, evaluation, and refinement.
- Creative work becomes iterative rather than linear.

Key message

AI transforms how designers work, not simply the tools they use.

2 Designer Adaptation



Technology alone is not enough

Successful AI adoption depends on designers':

- AI literacy
- willingness to experiment
- trust in AI
- maintaining creative control

Key message

AI adoption is primarily a human adaptation process.

3 Skills Development



The literature consistently shows the need for new competencies:

- To work effectively with AI tools
- To evaluate and refine AI-generated outputs
- To make ethical and responsible design decisions
- To collaborate with AI while sustaining originality

Key message

Future designers require both technical and creative competencies.

Takeaway

These three dimensions—workflow evolution, designer adaptation, and skills development—collectively explain the transition from CAD-enabled design to Human–AI co-creativity, forming the basis of the proposed conceptual framework.

Evolution of Fashion Design Practice

From Manual and CAD-Based Processes to AI-Supported Design

DIGITAL TRANSFORMATION

Evolution of Digital Technologies in Fashion Design Practice

MANUAL → CAD

(Digitization Era)

WORKFLOW CHANGES

- Linear → Digital
- Sequential process
- Enter editing & modification
- Faster drafting & visualization
- Focus on accuracy & efficiency

Sketch → Draft → Develop → Finalize

DESIGNER ADAPTATION

- Digital operator
- Digital drafter
- Technical designer
- Execute & develop ideas digitally
- Focus on producing accurate outputs

Role: Operator → Executor

SKILLS DEVELOPMENT

Technical Skills	Soft Skills
• CAD software	• Adaptability
• Digital illustration	• Digital confidence
• Technical drawing	• Attention to detail
• Pattern digitization	• Problem solving
• Digital rendering	• Time management

1. WORKFLOW CHANGES

How design processes evolve with digital technologies

2. DESIGNER ADAPTATION

How designers adapt their roles, responsibillities and mindsets

3. SKILLS DEVELOPMENT

How designers develop technical, soft and AI-related competencies

CAD → AI

(Intelligence Augmentation Era)

WORKFLOW CHANGES

- Digital → Iterative co-creation
- Non-linear, cyclical process
- Prompt → Generate → Evaluate → Refine → Re-prompt
- Continuous exploration
- Focus on creativity, evaluation & curation

Prompt → Generate → Evaluate → Refine → Integrate → Curate

DESIGNER ADAPTATION

- AI collaborator
- Curator & editor
- Creative strategist / director
- Evaluate, select & refine AI outputs
- Focus on meaning-making, value creation & originality

Role: Collaborator → Curator → Strategist

SKILLS DEVELOPMENT

Technical Skills	Soft Skills
• AI literacy	• Critical thinking
• Prompt engineering	• Creativity
• Generative AI tools	• Ethical judgment
• AI evaluation	• Collaboration
• Data interpretation	• Open-mindedness
• Workflow integration	• Continuous learning

Digitize & Document • Optimize & Collaborate • Co-create & Innovate

From digitization to augmentation – technology transforms tools, but **designers drive meaning, creativity and value.**

Conclusion

Key insights from the study and implications for the future.



AI integration is technological and human.

Successful adoption depends on both advanced tools and designer adaptation. Technology enables, but humans interpret, decide, and give meaning.

Human creativity remains central.

Designers bring context, emotion, cultural understanding, and intuition that AI cannot replicate. Creativity is amplified, not replaced.

Responsible Human–AI collaboration is the future of fashion design.

Ethical use, transparency, critical evaluation, and human oversight are essential to build meaningful, sustainable, and inclusive creative practices.

The Takeaway

- Designers are not replaced they are empowered.
- AI enhances efficiency, but human judgement shapes value.
- The future lies in balance between innovation and responsibility.
- Co-creativity leads to better designs and a better world.

Thank you